Nathan Huse

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Full-stack developer who ships production systems and the guardrails that keep them honest: policy-gated hardware writes, database-enforced access control, and test suites that fail before output can drift.

EXPERIENCE

Contract Developer, Solid Woodworx — 2026 - present

- Pilot pitched and accepted July 2026, merged to main, and running in beta on the shop floor
- Built the engine as a pure interpreter: every vehicle-specific fact (parts, rules, sheet layout, provenance) lives in a versioned YAML catalog, so adding a new kit is a new file and zero code
- 453-test pytest suite as of 2026-07-25, covering the catalog gates, the PDF and OCR intake parser, order resolution, and the golden-locked sheet renderer

Food Service Partner, Scripps Health, Mercy Hospital San Diego — Jul 2021 - Mar 2026

- Identified ordering bottlenecks during solo operations and built a full-stack ordering platform, deployed into production use by hospital staff
- 2025 Values in Action Honoree, Scripps Mercy Hospital San Diego

PROJECTS

Grow Tent Telemetry, Sole Developer — 2026 - present

- Hardware writes sit behind an explicit policy gate: an enable flag, an allowed-sources list, a per-minute rate limit, a runtime kill switch, and a protected-ports list that rejects any write to the outlet powering the Pi itself
- Reads the controller and power strip directly over Bluetooth LE by decoding the device's self-streaming packet format, with the vendor cloud reader removed from the codebase entirely
- A Raspberry Pi at the tent writes a remote Postgres system of record across a tailnet through a durable JSONL spool that buffers during network outages and backfills on recovery
- 849 tests in the root suite as of 2026-07-12, covering alert rules, BLE decode and control, the policy gate, MCP tools, and the edge proxy

CafeNightClub, Sole Developer — 2025 - 2026

- Built and shipped a production ordering system for 80+ night-shift hospital staff, with real-time order tracking over Supabase Realtime WebSockets
- Order creation runs as a single database RPC call, so there are no partial writes and no race between cart validation and inventory checks
- Row-Level Security policies on every table, with role-based access (user, admin, owner) enforced server-side rather than only in the UI, and rate limiting on cart operations

Arenula, Sole Developer — 2026

- Three servers split by runtime and transport: a C# plugin inside the s&box editor handles scene manipulation, compilation, assets and terrain over SSE, and two Node servers over stdio serve the offline type reference and the narrative docs
- Designed to Anthropic's published guidance on writing tools for agents: omnibus tools with action enums, descriptions carrying negative guidance, trimmed responses, and per-field partial-success warnings

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS
Frontend: React, Next.js, Astro, Tailwind CSS, DaisyUI, SSR, Responsive Design, Accessibility
Backend: Node.js, Bun, PostgreSQL, SQLite, Drizzle ORM, Supabase, better-auth, RESTful APIs, SSE, Docker, Linux (Ubuntu), Caddy, PM2
Tools & Practices: Git, GitHub, CI/CD (GitHub Actions), Vitest, pytest, AI-augmented development (Claude), State Management, Database Design
AI & Search: Semantic search, Vector embeddings and reranking (Voyage AI), pgvector (HNSW), MCP tool servers, Multi-agent orchestration

EDUCATION

San Diego State University, Bachelor of Science in Computer Science — 2020
Grossmont College, Associate of Science in Mathematics / Computer Science — 2017